

MPAS-URBAN QUICKSTART | GUIDE 1 OF 3

Install HKUST-MPAS

Start from the official HKUST-MPAS source, compile both atmospheric executables, and add the MPAS-Urban geographical data.

OUTCOME You will have a standard HKUST-MPAS installation for learning and new experiments, plus a clearly separated optional source tree for the HRAIN2025 NTDK cutoff experiment.

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Before you begin

This guide supplements rather than replaces the official MPAS-Atmosphere tutorial. New users should first complete the official tutorial or Quick Start Guide and confirm that a standard MPAS case can be compiled and run.

- Official tutorial: <https://mpas-dev.github.io/atmosphere/tutorial.html>
- Quick Start Guide:
https://www2.mmm.ucar.edu/projects/mpas/site/documentation/users_guide/quick_start.html

Use a Unix/Linux server. On an HPC system, prefer the compiler, MPI, NetCDF, and PnetCDF modules supported by the site. Keep one compiler/MPI family for MPAS and every MPI-linked library.

If the server does not yet provide a working MPAS build environment, first read the dependency sections of the official tutorial and then use Appendix A in this guide. Appendix A gives one complete user-space GNU/Open MPI example for a machine without an existing MPAS environment. On a managed HPC cluster, site-provided modules remain the preferred route.

Step 1 | Check the existing MPAS build environment

COMMAND

```
command -v gcc
command -v gfortran
command -v mpicc
command -v mpif90
command -v pnetcdf-config

mpicc --showme:command 2>/dev/null || mpicc -show
mpif90 --showme:command 2>/dev/null || mpif90 -show
```

CHECK For the gfortran build target, the MPI wrappers should resolve to a compatible GNU compiler family. Stop and correct the module environment if the wrappers point to a different compiler family.

ONE ROOT FOR THE COMPLETE TUTORIAL

All local source trees, dependencies, datasets, case files, and model output are kept under one root. The three guides use MPAS_ROOT so that new users do not have to manage unrelated directories across \$HOME.

```
$HOME/MPAS/
|-- HKUST-MPAS/
|-- HKUST-MPAS-NTDK/          # optional case-specific build
|-- DATA/mpas_static/
|   |-- mpas_cglc_lcz/
|-- HRAIN2025/
|-- HRAIN2025_INPUT/
|   |-- HRAIN2025_30km_500m/
```

Step 2 | Download the standard HKUST-MPAS source

Use the HKUST-MPAS team repository and its hkust-dev branch as the default source for MPAS-Urban learning and new independent experiments.

STANDARD SOURCE

```
export MPAS_ROOT="$HOME/MPAS"
mkdir -p "$MPAS_ROOT"
cd "$MPAS_ROOT"

git clone --depth 1 --branch hkust-dev --single-branch https://github.com/HKUST-MPAS/HKUST-MPAS.git

cd HKUST-MPAS
git branch --show-current
test -f Makefile
test -d src/core_atmosphere
```

EXPECTED `git branch --show-current` prints `hkust-dev`.

Step 3 | Optional HRAIN2025 NTDK cutoff source

Skip this step for a standard MPAS-Urban installation. The following repository is a case-author derivative used for the HRAIN2025 cutoff experiment; it is not the official HKUST-MPAS main branch and does not represent the default HKUST-MPAS physics configuration.

CASE-AUTHOR DERIVATIVE

```
export MPAS_ROOT="$HOME/MPAS"
cd "$MPAS_ROOT"

git clone --depth 1 --branch ntdk-grid-cutoff --single-branch https://github.com/Liuzh223/HKUST-MPAS-Official.git HKUST-MPAS-NTDK

cd HKUST-MPAS-NTDK
git branch --show-current
```

NAME NOTE The word `Official` is part of this GitHub repository name only. The `ntdk-grid-cutoff` branch is maintained for this case. There is no separate terminal command named `cutoff`.

OPTION NOTE This derivative adds `config_cu_disable_dx`. The value is selected later in Guide 3; it is not a compile command.

Step 4 | Compile both MPAS executables

Set PNETCDF to the prefix already prepared through the official tutorial or your cluster modules. Leave PIO unset to use the bundled SMIOOL path in this MPAS version.

BUILD ENVIRONMENT

```
# Example only: change this to the PnetCDF prefix on your system.
export MPAS_ROOT="$HOME/MPAS"
export PNETCDF="$MPAS_ROOT/opt/mpas-libs"
unset PIO

pnetcdf-config --version
```

Choose exactly one source directory before compiling:

```
# Standard HKUST-MPAS (default)
export MPAS_BUILD="$MPAS_ROOT/HKUST-MPAS"

# HRAIN2025 cutoff derivative (optional; use instead)
# export MPAS_BUILD="$MPAS_ROOT/HKUST-MPAS-NTDK"

cd "$MPAS_BUILD"
```

BUILD

```
make -j 4 gfortran CORE=init_atmosphere PRECISION=single
make clean CORE=atmosphere
make -j 1 gfortran CORE=atmosphere PRECISION=single
```

VERIFY

```
test -x init_atmosphere_model
test -x atmosphere_model
ls -lh init_atmosphere_model atmosphere_model
```

BUILD NOTE The serial atmosphere build avoids Fortran module races observed on some systems. Increase the parallel build count only after the toolchain has been verified.

Step 5 | Download the standard MPAS static data

DOWNLOAD

```
export MPAS_ROOT="$HOME/MPAS"
mkdir -p "$MPAS_ROOT/DATA"
cd "$MPAS_ROOT/DATA"

curl -fL -O https://www2.mmm.ucar.edu/projects/mpas/mpas_static.tar.bz2

tar -xjf mpas_static.tar.bz2
test -d "$MPAS_ROOT/DATA/mpas_static"
```

Use the absolute path to \$HOME/MPAS/DATA/mpas_static as config_geog_data_path during initialization.

Step 6 | Add the MPAS-Urban CGLC-LCZ data

DOWNLOAD AND EXTRACT

```
export MPAS_ROOT="$HOME/MPAS"
cd "$MPAS_ROOT/DATA"

curl -fL -O https://github.com/HKUST-MPAS/HKUST-
MPAS/releases/download/LCZ_dataset_for_MPAS_Urban/mpas_cglc_lcz.tar.gz

mkdir -p "$MPAS_ROOT/DATA/mpas_static/mpas_cglc_lcz"
tar -xzf mpas_cglc_lcz.tar.gz -C "$MPAS_ROOT/DATA/mpas_static/mpas_cglc_lcz"

test -s "$MPAS_ROOT/DATA/mpas_static/mpas_cglc_lcz/index"
```

The HKUST-MPAS source already supplies the MPAS-Urban parameter tables. Do not download replacement urban tables unless a specific code version instructs you to do so.

Ready for Guide 2

- `init_atmosphere_model` and `atmosphere_model` exist.
- The standard MPAS static dataset is available.
- `mpas_static/mpas_cglc_lcz/index` exists.

Appendix A | Build the MPAS dependencies from scratch

Use this appendix only when the server has no compatible MPI, NetCDF, or PnetCDF environment. Install all user-built libraries under one prefix and use the same GNU compiler and Open MPI wrappers for every MPI-linked package.

A.1 | Example versions

- Open MPI 4.1.6
- zlib 1.3.1
- HDF5 1.14.6
- NetCDF-C 4.9.3
- NetCDF-Fortran 4.6.4
- PnetCDF 1.12.3

These are pinned tutorial examples, not a requirement to use the newest release. Changing several versions at once makes failures harder to diagnose.

A.2 | Create the local prefix

```
export MPAS_ROOT="$HOME/MPAS"
export MPAS_PREFIX="$MPAS_ROOT/opt/mpas-libs"
export MPAS_LIB_SRC="$MPAS_ROOT/src/mpas-libs"
mkdir -p "$MPAS_PREFIX" "$MPAS_LIB_SRC"

export PATH="$MPAS_PREFIX/bin:$PATH"
export LD_LIBRARY_PATH="$MPAS_PREFIX/lib:$MPAS_PREFIX/lib64"
export CPATH="$MPAS_PREFIX/include"
export LIBRARY_PATH="$MPAS_PREFIX/lib:$MPAS_PREFIX/lib64"
export PKG_CONFIG_PATH="$MPAS_PREFIX/lib/pkgconfig:$MPAS_PREFIX/lib64/pkgconfig"
export CPPFLAGS="-I$MPAS_PREFIX/include"
export LDFLAGS="-L$MPAS_PREFIX/lib -L$MPAS_PREFIX/lib64"
```

A.3 | Optional user-space Open MPI

Skip this section when compatible mpicc, mpif90, and mpiexec commands already exist.

```
export MPAS_ROOT="$HOME/MPAS"
export MPI_PREFIX="$MPAS_ROOT/opt/openmpi-4.1.6"
mkdir -p "$MPAS_ROOT/src" "$MPI_PREFIX"
cd "$MPAS_ROOT/src"

curl -fL -O https://download.open-mpi.org/release/open-mpi/v4.1/openmpi-4.1.6.tar.gz
tar -xzf openmpi-4.1.6.tar.gz
cd openmpi-4.1.6

./configure --prefix="$MPI_PREFIX" CC=gcc CXX=g++ FC=gfortran
make -j 8
make install

export PATH="$MPI_PREFIX/bin:$PATH"
export LD_LIBRARY_PATH="$MPI_PREFIX/lib:$LD_LIBRARY_PATH"
mpicc --showme:command
mpif90 --showme:command
```

A.4 | Build zlib

```
cd "$MPAS_LIB_SRC"
curl -fL -O https://zlib.net/fossils/zlib-1.3.1.tar.gz
tar -xzf zlib-1.3.1.tar.gz
cd zlib-1.3.1
./configure --prefix="$MPAS_PREFIX"
make -j 8
make check
make install
```

A.5 | Build HDF5

```
cd "$MPAS_LIB_SRC"
curl -fL -O https://github.com/HDFGroup/hdf5/releases/download/hdf5_1.14.6/hdf5-1.14.6.tar.gz
tar -xzf hdf5-1.14.6.tar.gz
cd hdf5-1.14.6
CC=gcc ./configure --prefix="$MPAS_PREFIX" --with-zlib="$MPAS_PREFIX" --enable-shared --
disable-fortran
make -j 8
make check
make install
```

A.6 | Build NetCDF-C

```
cd "$MPAS_LIB_SRC"
curl -fL -O https://downloads.unidata.ucar.edu/netcdf-c/4.9.3/netcdf-c-4.9.3.tar.gz
tar -xzf netcdf-c-4.9.3.tar.gz
cd netcdf-c-4.9.3
CC=gcc ./configure --prefix="$MPAS_PREFIX" --disable-dap
make -j 8
make check
make install
nc-config --version
nc-config --has-nc4
```

A.7 | Build NetCDF-Fortran

```
cd "$MPAS_LIB_SRC"
curl -fL -O https://downloads.unidata.ucar.edu/netcdf-fortran/4.6.4/netcdf-fortran-4.6.4.tar.gz
tar -xzf netcdf-fortran-4.6.4.tar.gz
cd netcdf-fortran-4.6.4
CC=gcc FC=gfortran ./configure --prefix="$MPAS_PREFIX"
make -j 8
make check
make install
nf-config --version
```

A.8 | Build PnetCDF with the selected MPI wrappers

```
cd "$MPAS_LIB_SRC"
curl -fL -O https://parallel-netcdf.github.io/Release/pnetcdf-1.12.3.tar.gz
tar -xzf pnetcdf-1.12.3.tar.gz
cd pnetcdf-1.12.3
CC=mpicc FC=mpif90 ./configure --prefix="$MPAS_PREFIX"
make -j 8
make check
make install
pnetcdf-config --version
```

A.9 | Export the MPAS build variables

```
export PATH="$MPAS_PREFIX/bin:$PATH"
export LD_LIBRARY_PATH="$MPAS_PREFIX/lib:$MPAS_PREFIX/lib64:$LD_LIBRARY_PATH"
unset NETCDF NETCDFF PNETCDF PIO
export PNETCDF="$MPAS_PREFIX"
```

```
which mpicc mpif90
pnetcdf-config --version
nc-config --version
nf-config --version
```

Return to Step 2 after these checks pass. If a make check command fails, do not continue to MPAS compilation until the compiler, MPI wrapper, header, and library paths are consistent.